Lectures 19 – 21

Math 4741: Mathematical Analysis

Chapter 8: Hidden Markov Models

References

Texts: Markov Processes for Stochastic Modeling – Oliver Ibe (Chapter 14)

Hidden Markov Models (HMMs)

HMM Basics:

An HMM is usually defined as a 5-tuple $(S, \Omega, P, \Phi, \pi)$

$S =$

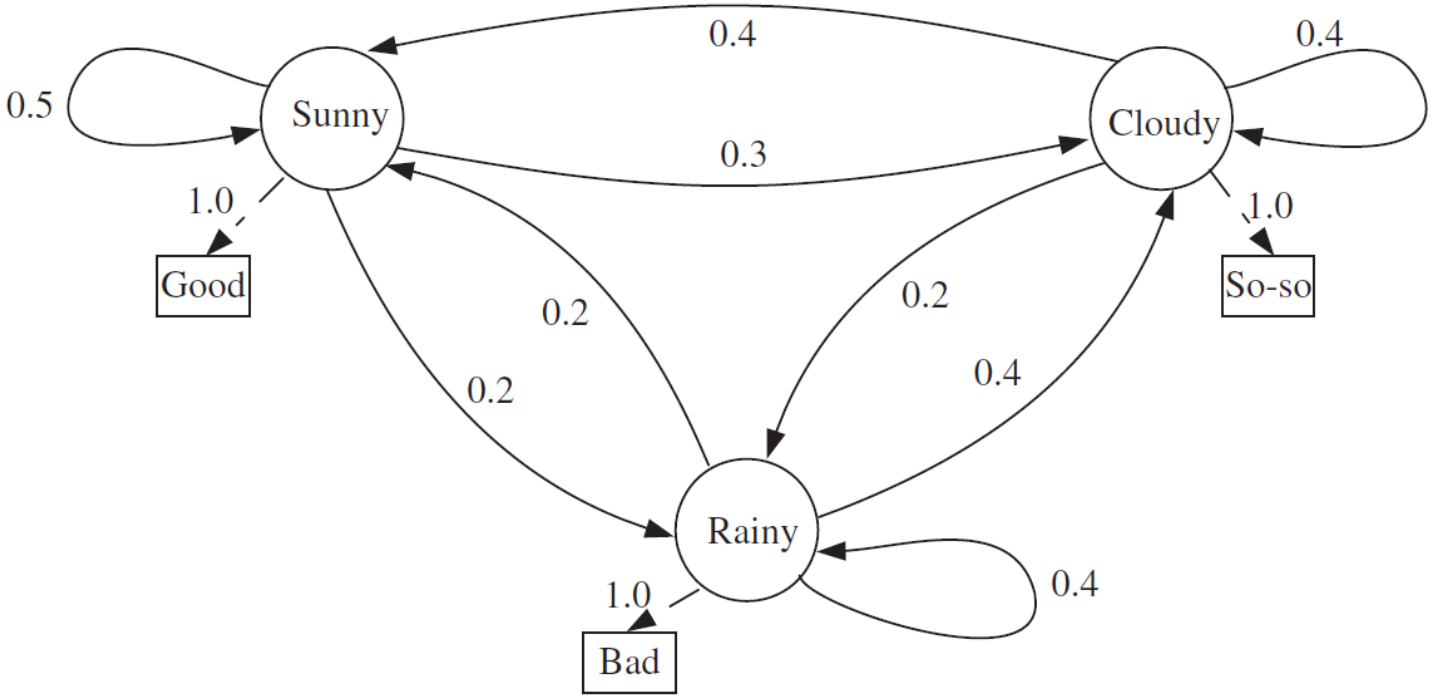
$\Omega =$

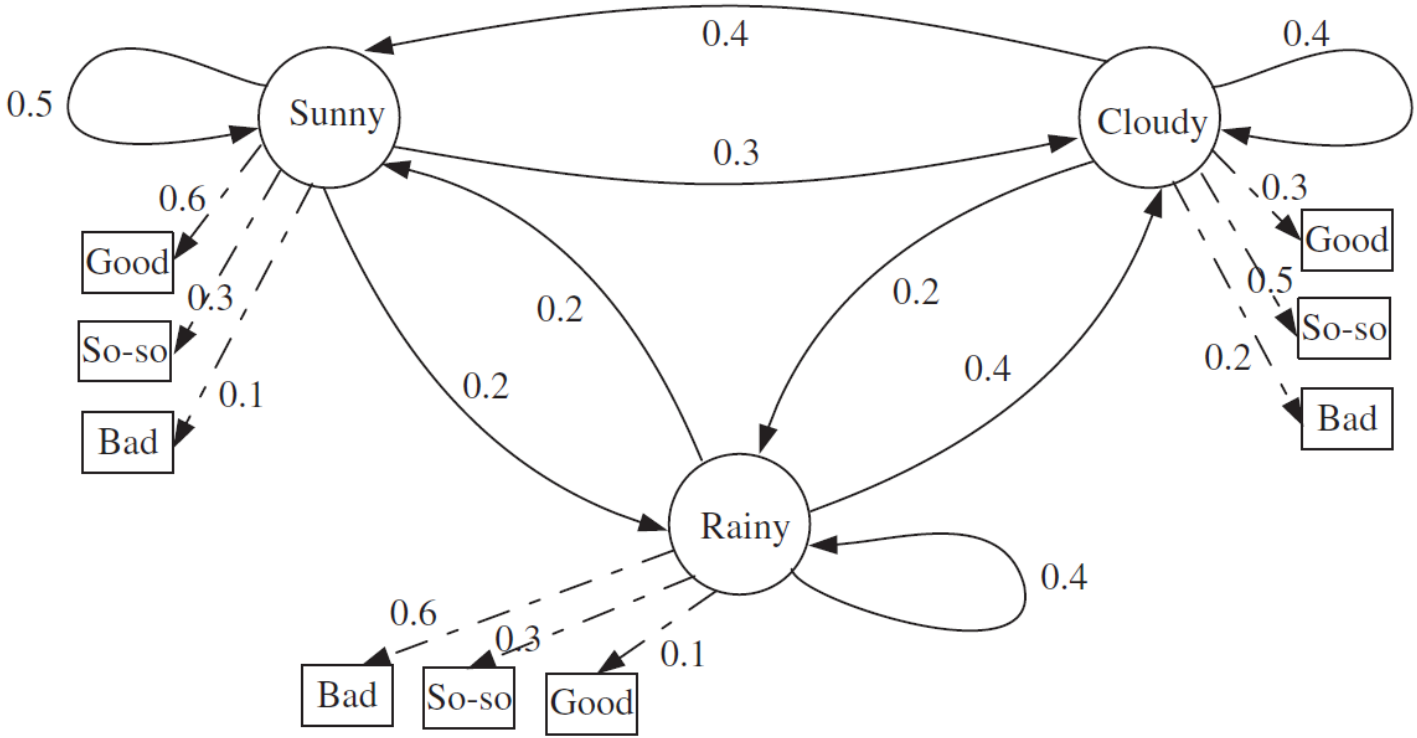
$P =$

$\Phi =$

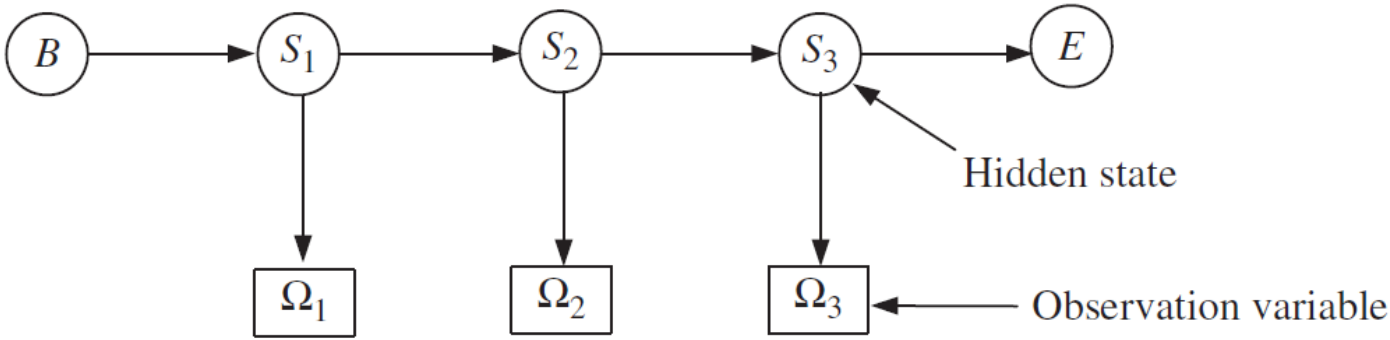
$\pi =$

Parameters of an HMM


0.4
0.4
0.5
Sunny
Cloudy
0.3
1.0
1.0
Good
So-so
0.2
0.2
0.2
0.4
Rainy
0.4
1.0
Bad


Sunny
Cloudy
Rainy
0.5
0.4
0.4
0.3
0.6
0.3
0.1
0.2
0.2
0.2
0.4
0.3
0.5
0.2
0.4
0.6
0.3
0.1
Good
So-so
Bad
Good
So-so
Bad
Bad
So-so
Good

General Structure of an HMM



HMM Assumptions

1. Markov assumptions

2. Stationary assumptions

3. Observation independence assumptions

Three Fundamental Problem

1. *The evaluation problem*: Given a model $\lambda = (P, \Phi, \pi)$ and an observation sequence $O = v_1, v_2, \ldots, v_T$ of length T, where $v_i \in \Omega$, how do we efficiently compute the probability that the model generated the observation sequence; that is, what is $P[O|\lambda]$?

2. *The decoding problem*: Given a model $\lambda = (P, \Phi, \pi)$, what is the most likely sequence of hidden states that could have generated a given observation sequence? Thus, we would like to find $Q^* = \arg\max_Q P[Q, O|\lambda]$, where Q is the hidden state sequence, as defined earlier.

3. *The learning problem*: Given a set of observation sequences, find the HMM that best explains the observation sequences; that is, find the values for λ that maximize $P[O|\lambda]$ or $\lambda^* = \arg\max_\lambda P[O|\lambda]$. Stated differently, the problem is to estimate the most likely HMM parameters for a given observation sequence.

Solution Methods

1. Evaluation Problem: Forward algorithm and backward algorithm

2. Decoding problem: Viterbi algorithm

3. Learning Problem: Baum-Welch algorithm

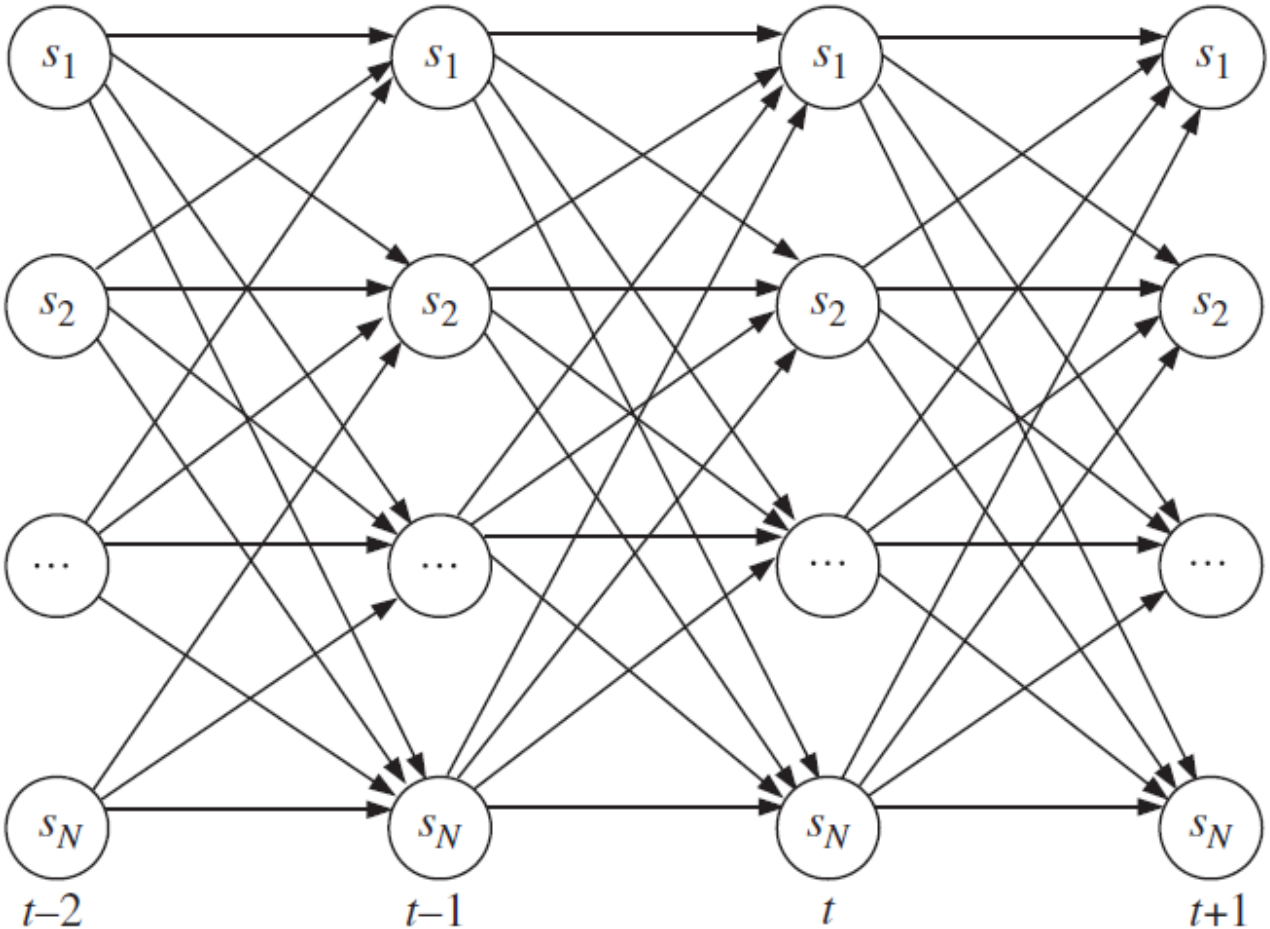
The Evaluation method

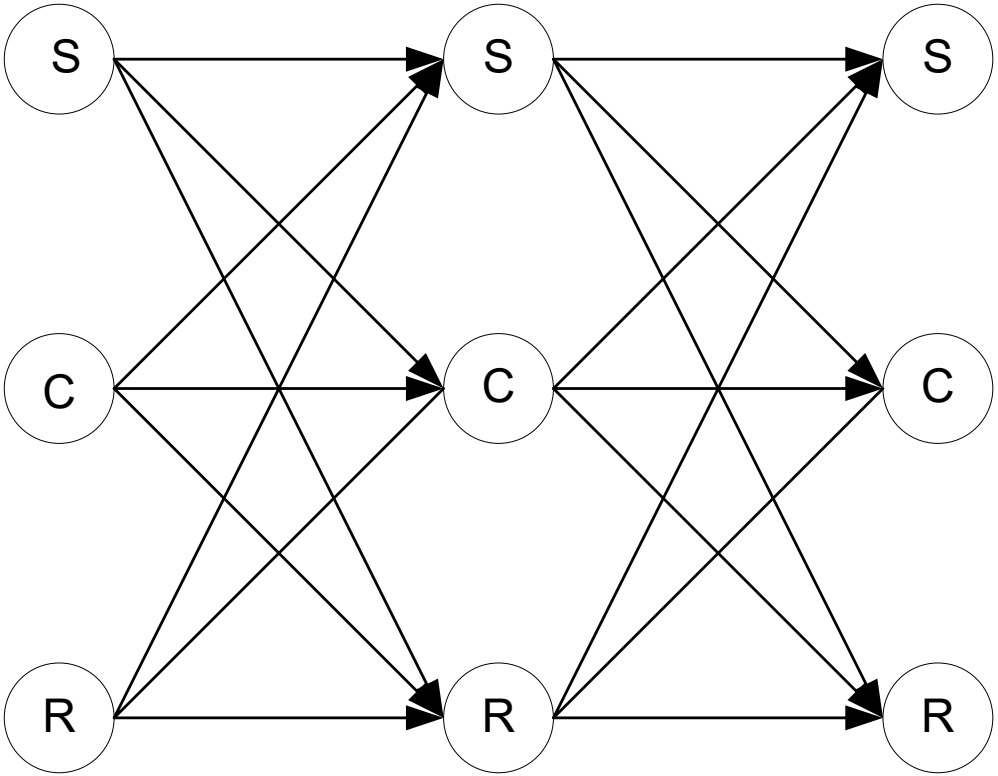
The forward Algorithm

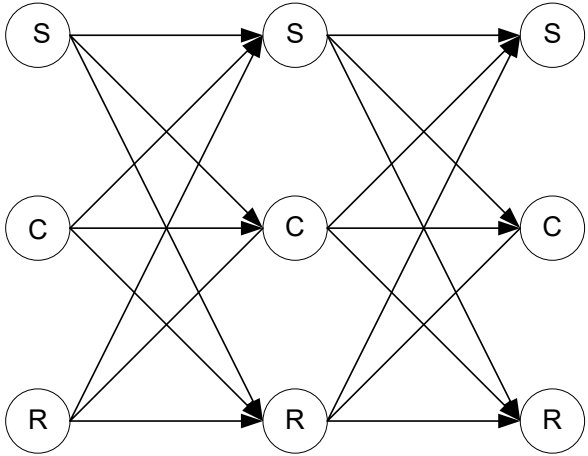
Find the probability of all paths that end in a particular node, where for a path

- Probability of sequence of nodes visited in the path, and

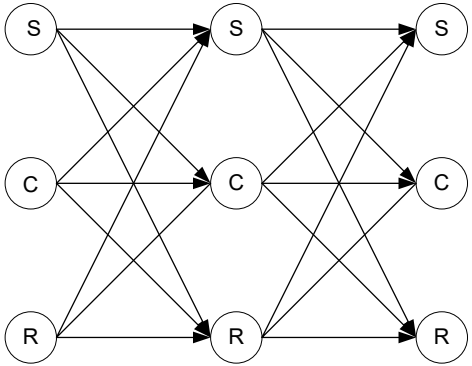
- Probability of sequence of symbols emitted in the path


s_1
s_2
…
s_N
$t-2$
$t-1$
t
$t+1$


S
C
R
S
C
R
S
C
R


S
C
R
S
C
R
S
C
R

A forward probability variable $\alpha_t(i)$ is used:



The forward algorithm

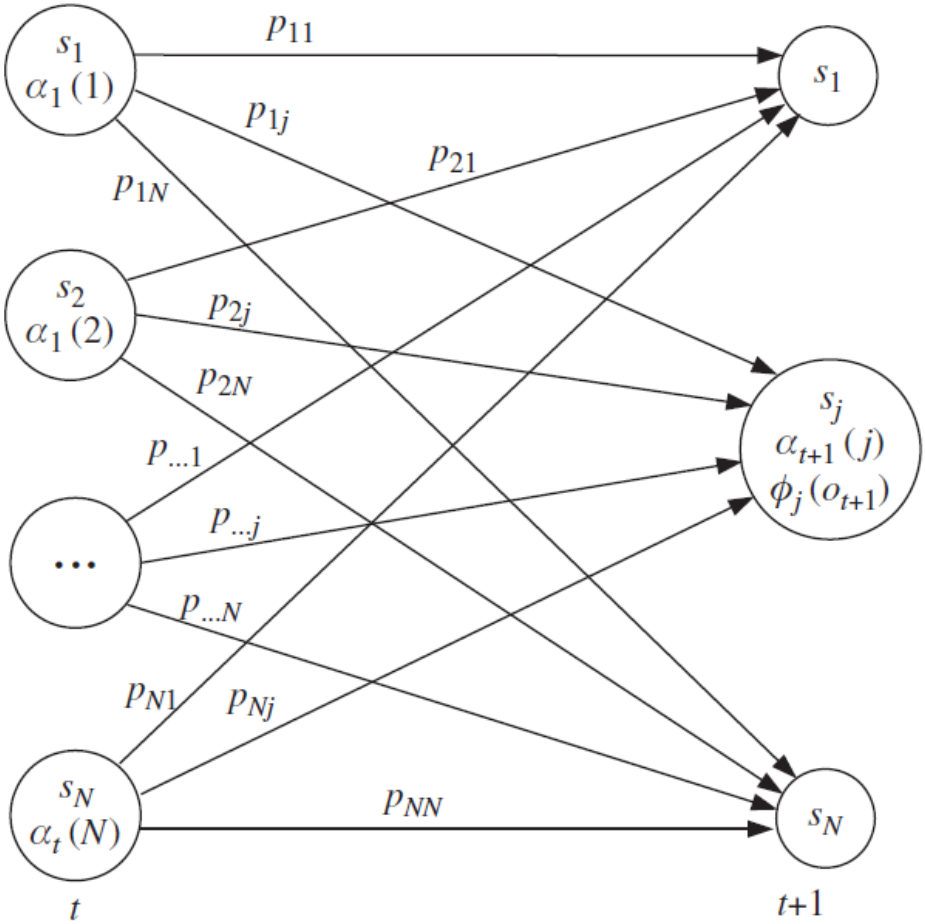
1. Initialization

2. Induction

3. Update

4. Termination

The forward algorithm requires $N(N+1)(T-1)+N$ multiplications and $N(N-1)(T-1)$ additions, giving a complexity on the order of N^2T rather than $2TN^T$.


s_1
$\alpha_1(1)$
s_2
$\alpha_1(2)$
...
s_N
$\alpha_t(N)$
t
p_{11}
p_{1j}
p_{1N}
p_{21}
p_{2j}
p_{2N}
$p_{...1}$
$p_{...j}$
$p_{...N}$
p_{N1}
p_{Nj}
p_{NN}
s_1
s_j
$\alpha_{t+1}(j)$
$\phi_j(o_{t+1})$
s_N
$t+1$

Example 14.1

Consider Bill's mood change problem illustrated in Figure 13.2. Assume that we observed Bill in the following sequence of moods: *Good*, *Good*, *So-so*, *Bad*, *Bad*. We are required to find the probability that the model generated such a sequence of moods using the forward algorithm.

Solution

Let S denotes Sunny state, C denotes Cloudy, and R denotes Rainy. Similarly, let G denotes Good mood, SS denotes So-so mood, and B denotes Bad mood. We assume that the process is equally likely to start in any state. That is, we assume that $\pi_S = \pi_C = \pi_R = 1/3$. Also, we have that $T = 5$. Then the initialization step becomes

Given

$$P = \begin{bmatrix} .5 & .3 & .2 \\ .4 & .4 & .2 \\ .2 & .4 & .4 \end{bmatrix} \quad \Phi = \begin{bmatrix} .6 & .3 & .1 \\ .3 & .5 & .2 \\ .1 & .3 & .6 \end{bmatrix} \quad \pi = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \quad O = \{G, G, S, B, B\}$$

1. Initialization step

2. Induction step $t = 2$

The induction step for $t = 2$ is given by

$$\alpha_2(j) = \left\{ \sum_{i=1}^{N} p_{ij}\alpha_1(i) \right\} \phi_j(o_2) = \left\{ \sum_{i=1}^{N} p_{ij}\alpha_1(i) \right\} \phi_j(G)$$

$$\begin{aligned}
\alpha_2(S) &= \{p_{SS}\alpha_1(S) + p_{CS}\alpha_1(C) + p_{RS}\alpha_1(R)\}\phi_S(G) \\
&= \{(0.5)(0.2) + (0.4)(0.1) + (0.2)(0.033)\}(0.6) \\
&= 0.088 \\
\alpha_2(C) &= \{p_{SC}\alpha_1(S) + p_{CC}\alpha_1(C) + p_{RC}\alpha_1(R)\}\phi_C(G) \\
&= \{(0.3)(0.2) + (0.4)(0.1) + (0.4)(0.033)\}(0.3) \\
&= 0.034 \\
\alpha_2(R) &= \{p_{SR}\alpha_1(S) + p_{CR}\alpha_1(C) + p_{RR}\alpha_1(R)\}\phi_R(G) \\
&= \{(0.2)(0.2) + (0.2)(0.1) + (0.4)(0.033)\}(0.1) \\
&= 0.007
\end{aligned}$$

$$\alpha_5(j) = \left\{ \sum_{i=1}^{N} p_{ij}\alpha_4(i) \right\} \phi_j(o_5) = \left\{ \sum_{i=1}^{N} p_{ij}\alpha_4(i) \right\} \phi_j(B)$$

$$\begin{aligned}
\alpha_5(S) &= \{p_{SS}\alpha_4(S) + p_{CS}\alpha_4(C) + p_{RS}\alpha_4(R)\}\phi_S(B) \\
&= \{(0.5)(0.002) + (0.4)(0.003) + (0.2)(0.007)\}(0.1) \\
&= 0.0004 \\
\alpha_5(C) &= \{p_{SC}\alpha_4(S) + p_{CC}\alpha_4(C) + p_{RC}\alpha_4(R)\}\phi_C(B) \\
&= \{(0.3)(0.002) + (0.4)(0.003) + (0.4)(0.007)\}(0.2) \\
&= 0.0009 \\
\alpha_5(R) &= \{p_{SR}\alpha_4(S) + p_{CR}\alpha_4(C) + p_{RR}\alpha_4(R)\}\phi_R(B) \\
&= \{(0.2)(0.002) + (0.2)(0.003) + (0.4)(0.007)\}(0.6) \\
&= 0.0023
\end{aligned}$$

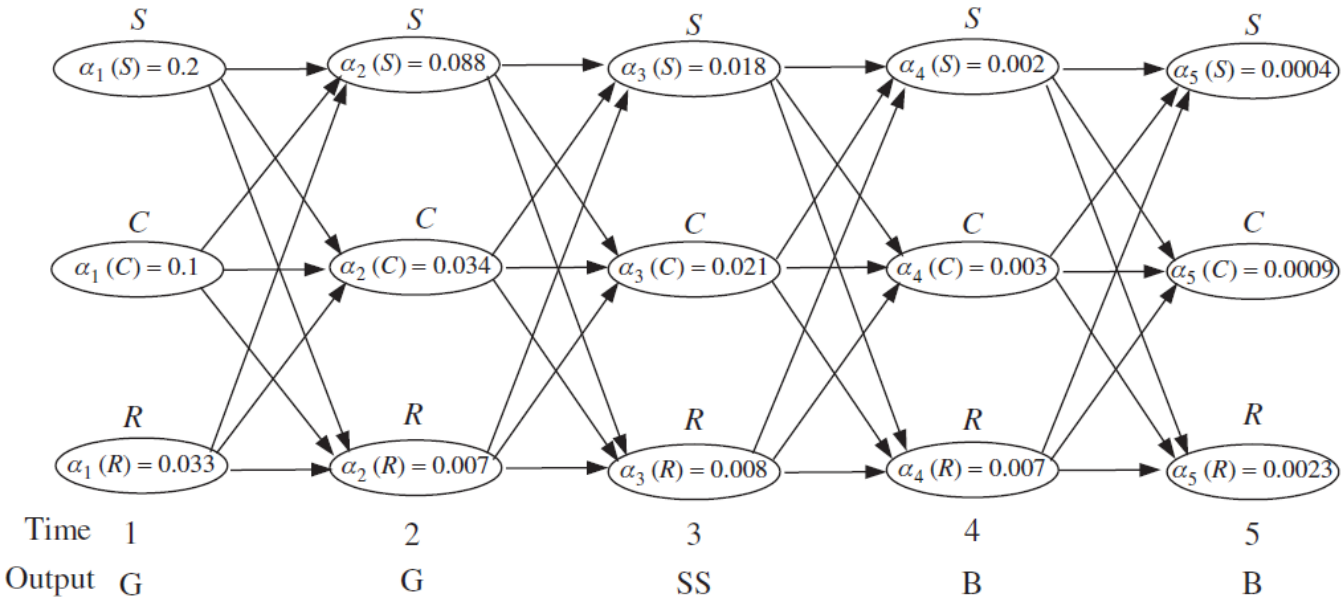
$$P[O = G, G, SS, B, B|\lambda] = \sum_{i=1}^{N} \alpha_T(i) = \alpha_T(S) + \alpha_T(C) + \alpha_T(R) = 0.0036$$

Forward Algorithm in the Matrix Form

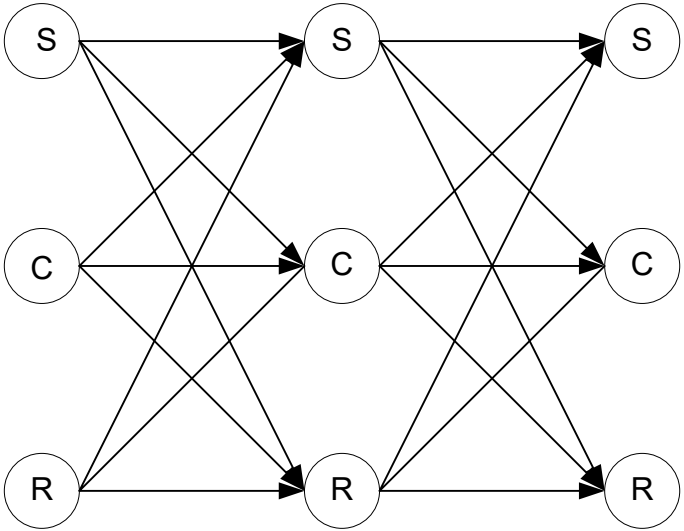
Initialization step

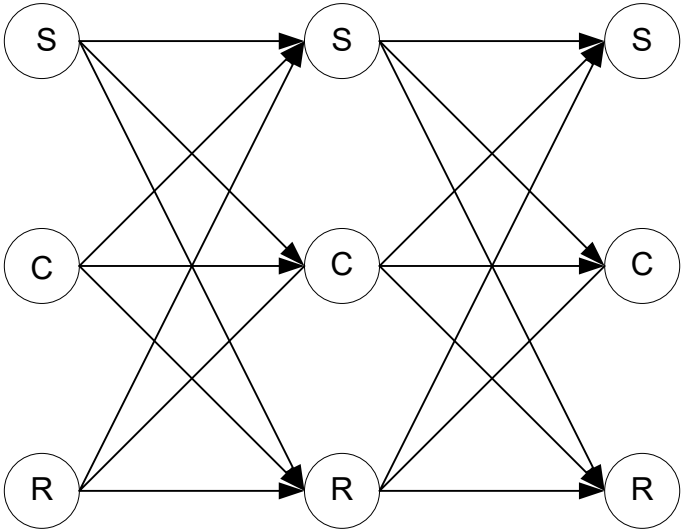
Step 2:

Step 5:


S
C
R
$\alpha_1(S) = 0.2$
$\alpha_1(C) = 0.1$
$\alpha_1(R) = 0.033$
$\alpha_2(S) = 0.088$
$\alpha_2(C) = 0.034$
$\alpha_2(R) = 0.007$
$\alpha_3(S) = 0.018$
$\alpha_3(C) = 0.021$
$\alpha_3(R) = 0.008$
$\alpha_4(S) = 0.002$
$\alpha_4(C) = 0.003$
$\alpha_4(R) = 0.007$
$\alpha_5(S) = 0.0004$
$\alpha_5(C) = 0.0009$
$\alpha_5(R) = 0.0023$
Time 1 2 3 4 5
Output G G SS B B

The Backward Algorithm


S
C
R
S
C
R
S
C
R


S
C
R
S
C
R
S
C
R

Initialization

Induction

Update

Termination

$$\beta_5(S) = \beta_5(C) = \beta_5(R) = 1$$

The induction step for $t = 4$ is given by

$$\beta_4(i) = \sum_{j=1}^{N} p_{ij}\beta_5(j)\phi_j(o_5)$$

$$\begin{aligned}
\beta_4(S) &= p_{SS}\beta_5(S)\phi_S(o_5) + p_{SC}\beta_5(C)\phi_C(o_5) + p_{SR}\beta_5(R)\phi_R(o_5) \\
&= p_{SS}\beta_5(S)\phi_S(B) + p_{SC}\beta_5(C)\phi_C(B) + p_{SR}\beta_5(R)\phi_R(B) \\
&= (0.5)(1)(0.1) + (0.3)(1)(0.2) + (0.2)(1)(0.6) = 0.23 \\
\beta_4(C) &= p_{CS}\beta_5(S)\phi_S(o_5) + p_{CC}\beta_5(C)\phi_C(o_5) + p_{CR}\beta_5(R)\phi_R(o_5) \\
&= p_{CS}\beta_5(S)\phi_S(B) + p_{CC}\beta_5(C)\phi_C(B) + p_{CR}\beta_5(R)\phi_R(B) \\
&= (0.4)(1)(0.1) + (0.4)(1)(0.2) + (0.2)(1)(0.6) = 0.24 \\
\beta_4(R) &= p_{RS}\beta_5(S)\phi_S(o_5) + p_{RC}\beta_5(C)\phi_C(o_5) + p_{RR}\beta_5(R)\phi_R(o_5) \\
&= p_{RS}\beta_5(S)\phi_S(B) + p_{RC}\beta_5(C)\phi_C(B) + p_{RR}\beta_5(R)\phi_R(B) \\
&= (0.2)(1)(0.1) + (0.4)(1)(0.2) + (0.4)(1)(0.6) = 0.34
\end{aligned}$$

The Decoding Problem: The Viterbi Algorithm

$$\arg\max_{Q} \{P[Q|O,\lambda]\} = \arg\max_{Q} \left\{\frac{P[Q,O|\lambda]}{P[O|\lambda]}\right\} = \arg\max_{Q} \{P[Q,O|\lambda]\}$$

$$\delta_t(i) = \max_{q_1,q_2,\ldots,q_{t-1}} P[q_1,q_2,\ldots,q_{t-1},q_t = s_i, o_1,o_2,\ldots,o_{t-1},o_t|\lambda]$$

The Viterbi Algorithm

1. Initialization:

$$\delta_1(i) = \pi_i \phi_i(o_1)$$
$$\psi_1(i) = 0 \qquad\qquad 1 \le i \le N$$

2. Recursion:

$$\delta_t(j) = \max_{1 \le i \le N} \{\delta_{t-1}(i) p_{ij}\} \phi_j(o_t)$$
$$\psi_t(j) = \arg \max_{1 \le i \le N} \{\delta_{t-1}(i) p_{ij}\} \quad 1 \le j \le N, \;\; 2 \le t \le T$$

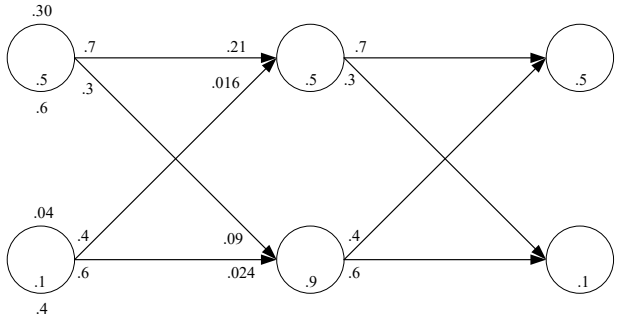
3. Termination

$$P^* = \max_{1 \le i \le N} \{\delta_T(i)\}$$
$$q_T^* = \arg \max_{1 \le i \le N} \{\delta_T(i)\}$$

Example:

$S = \{1, 2\}$ $\Omega = \{x, y\}$ $\pi = [0.6 \;\; 0.4]^T$ $O = \{x, y, x\}$

$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$ $\Phi = \begin{bmatrix} 0.\;5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix}$ $\Phi(x) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.1 \end{bmatrix}$ $\Phi(y) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.9 \end{bmatrix}$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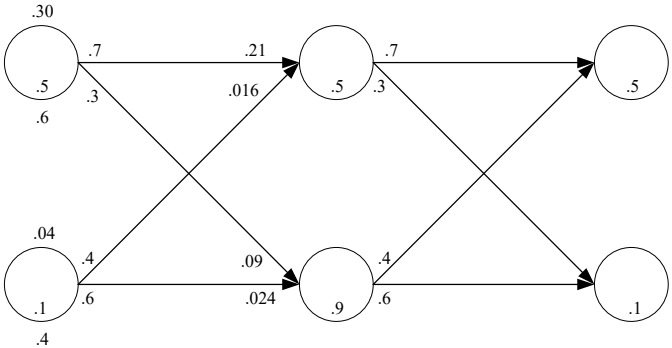
.30
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.1
.6
.024
.9
.6
.1
.4

The Learning Problem: Baum-Welch Algorithm

Given, $O = \{o_1, o_2, \dots, o_T\}$ and $\lambda = (P, \Phi, \pi)$

Find the best $\lambda = (P, \Phi, \pi)$ that describe the HMM

$$\lambda^* = \arg \max_{\lambda} \{P[O|\lambda]\}$$

$$\gamma_t(i) = \frac{\alpha_t(i)\beta_t(i)}{P[O|\lambda]} = \frac{\alpha_t(i)\beta_t(i)}{\sum_{i=1}^{N} \beta_t(i)\alpha_t(i)}$$

$$\xi_t(i,j) = P\left[q_t = s_i, q_{t+1} = s_j | O, \lambda\right] = \frac{P[q_t = s_i, q_{t+1} = s_j, O|\lambda]}{P[O|\lambda]}$$

$$= \frac{\alpha_t(i) p_{ij} \phi_j(o_{t+1}) \beta_{t+1}(j)}{\sum_{i=1}^{N} \beta_t(i) \alpha_t(i)} = \frac{\alpha_t(i) p_{ij} \phi_j(o_{t+1}) \beta_{t+1}(j)}{\sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_t(i) p_{ij} \phi_j(o_{t+1}) \beta_{t+1}(i)}$$

Now, the learning problem becomes, given the follwoings

1. Existing model $\lambda = (P, \Phi, \pi)$
2. Set of observations, O
3. Set of tools (known variables) $\alpha_t(i)$, $\beta_t(i)$, $\gamma_t(i)$ and $\xi_t(i,j)$

Find $\bar{\lambda} = ?$

1. Find the initial state probability $\bar{\pi} = \gamma_1(i)$
2. $$\bar{p}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}$$
3. $$\bar{\phi}_j(k) = \frac{\sum_{t=1, o_t=k}^{T} \gamma_t(j)}{\sum_{t=1}^{T} \gamma_t(j)}$$

1. Obtain the estimate of the initial state distribution for state s_i as the expected frequency with which state is visited at time $t = 1$; that is

$$\overline{\pi}_i = \gamma_1(i)$$

2. Obtain the estimates $\overline{p}_{ij}$ and $\overline{\phi}_j(k)$ as defined earlier.
3. Let the current model be $\lambda = (P, \Phi, \pi)$ that is used to compute the values of $\overline{p}_{ij}$ and $\overline{\phi}_j(k)$. Let the reestimated model be $\overline{\lambda} = (\overline{P}, \overline{\Phi}, \overline{\pi})$. Using the updated model $\overline{\lambda} = (\overline{P}, \overline{\Phi}, \overline{\pi})$, we perform a new iteration until convergence.
4. If $P[O|\overline{\lambda}] - P[O|\lambda] < \varepsilon$, stop, where ε is a predefined threshold value.

The EM theory states that after each iteration, one of two things can happen:

a. $\overline{\lambda}$ is more likely than λ in the sense that $P[O|\overline{\lambda}] > P[O|\lambda]$ or
b. we have reached a stationary point of the likelihood function at which $\overline{\lambda} = \lambda$.